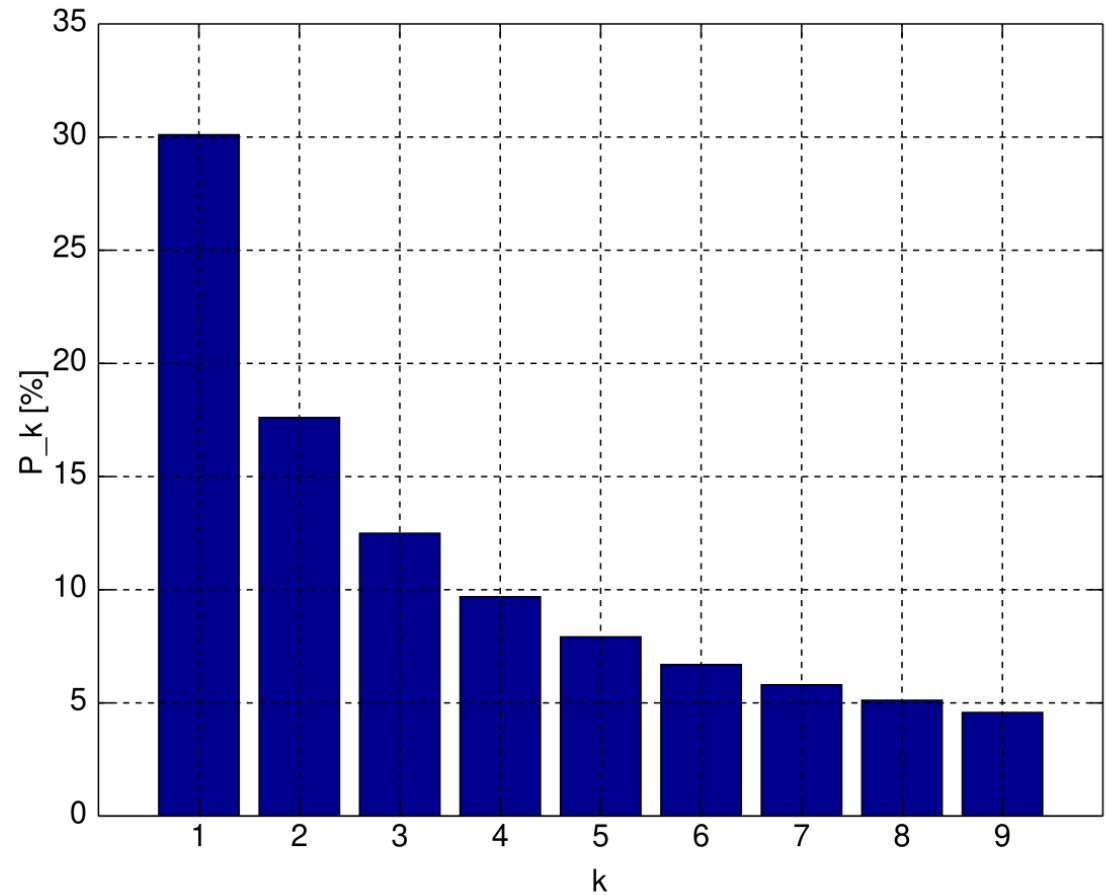


SPRITE

Benford's Law

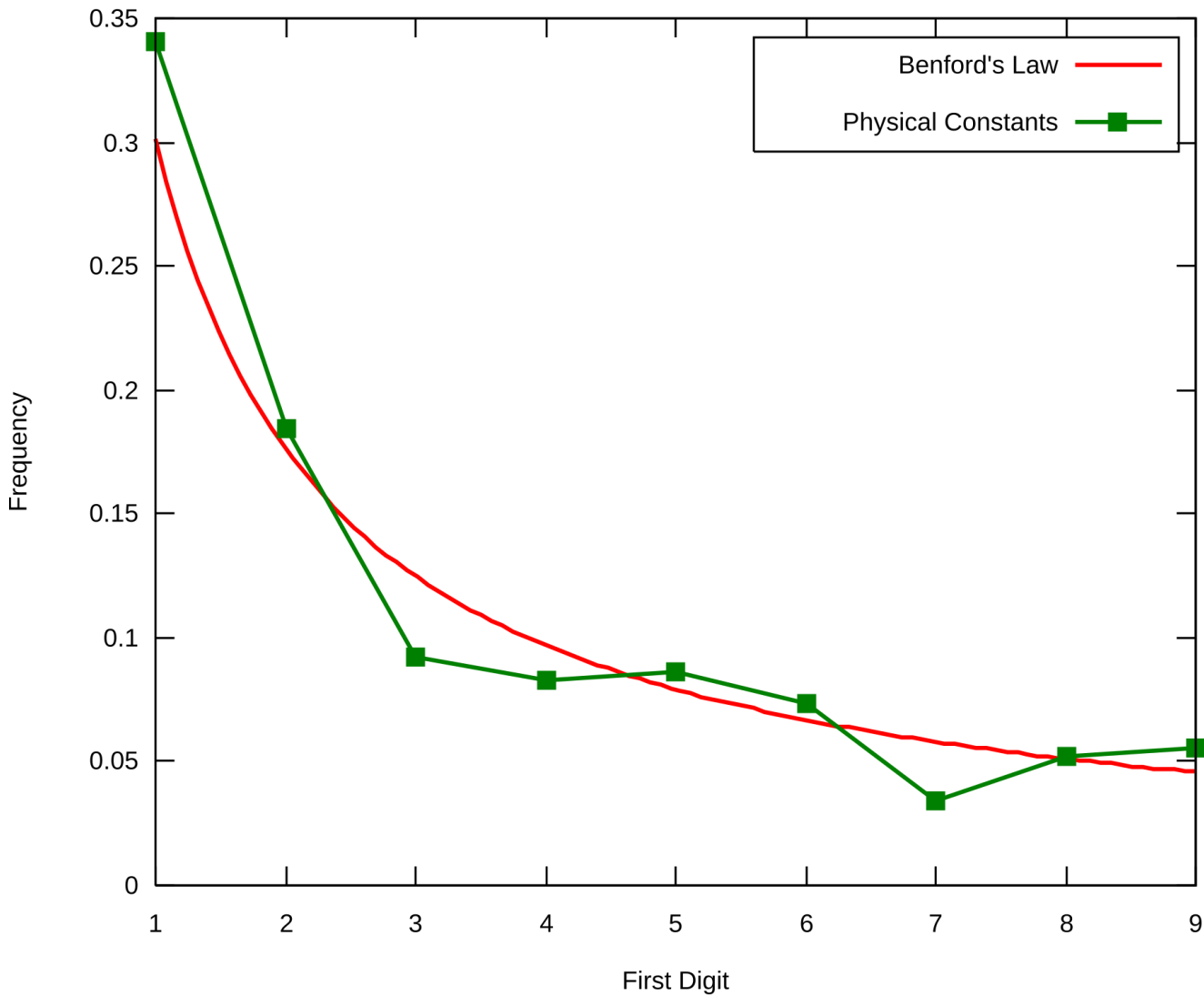
A set of numbers is said to satisfy Benford's law if the leading digit d ($d \in \{1, \dots, 9\}$) occurs with probability:

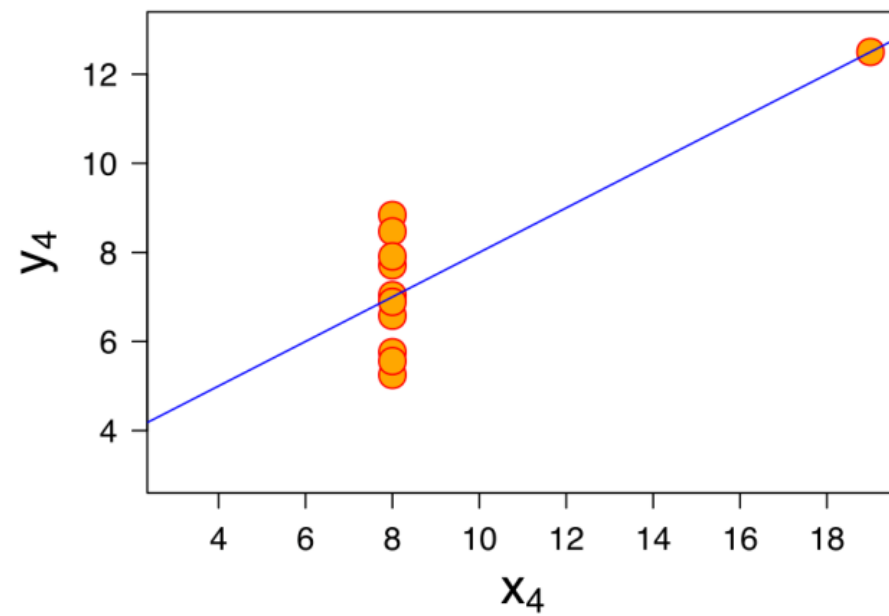
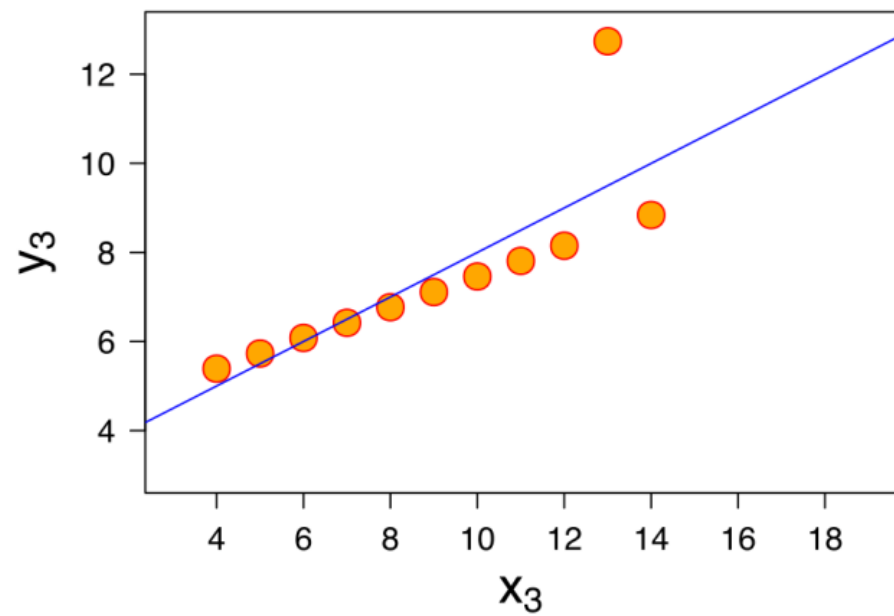
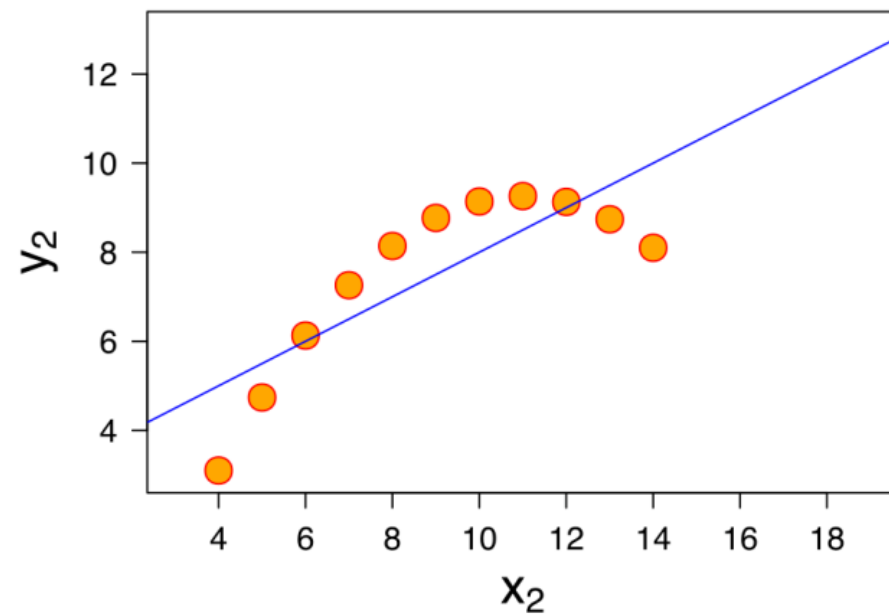
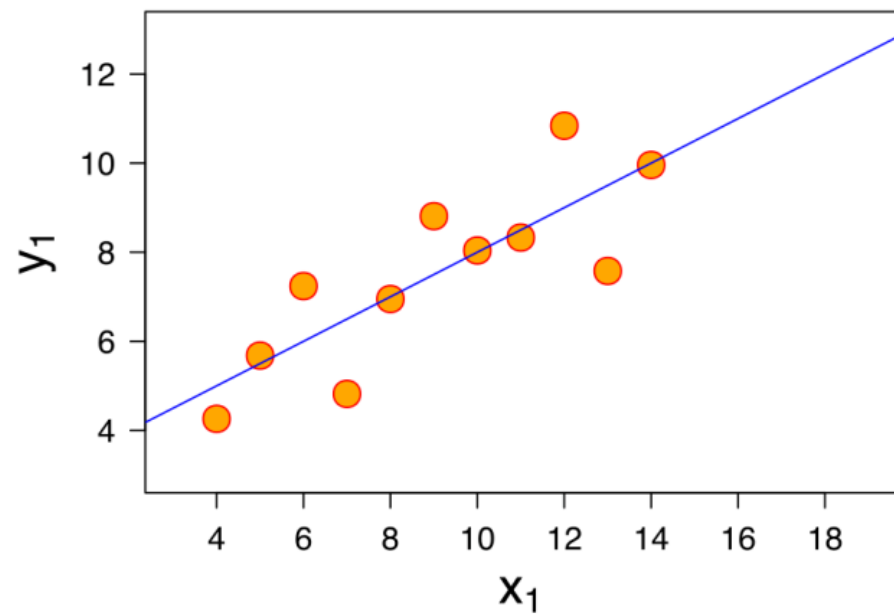


$$P(d) = \log_{10}(d+1) - \log_{10}(d) = \log_{10}\left(\frac{d+1}{d}\right) = \log_{10}\left(1 + \frac{1}{d}\right).$$

Forensic Accounting Application

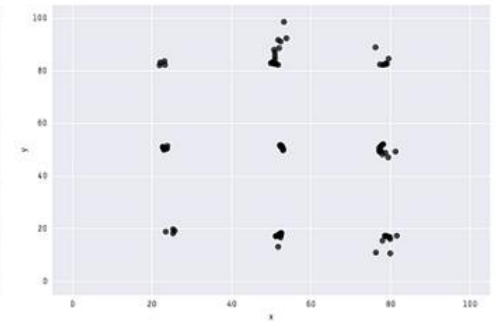
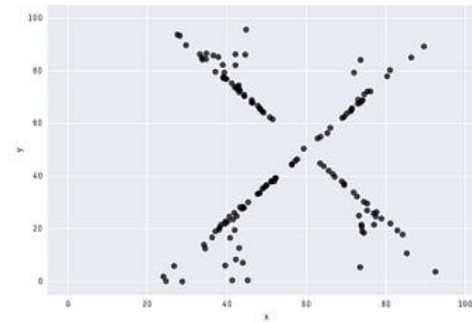
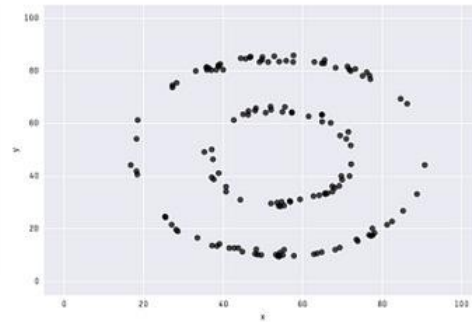
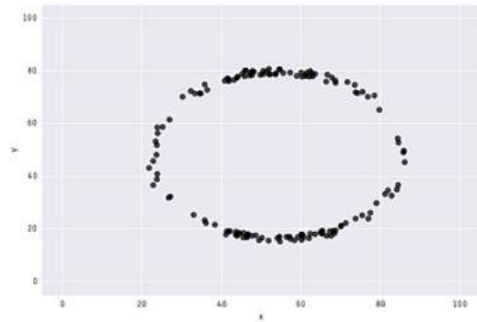
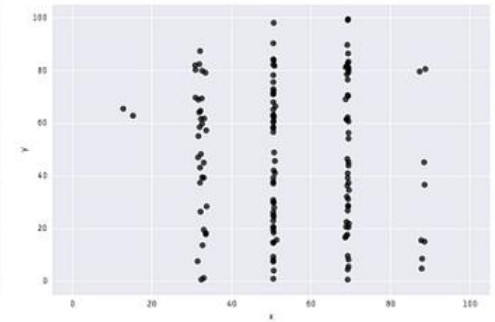
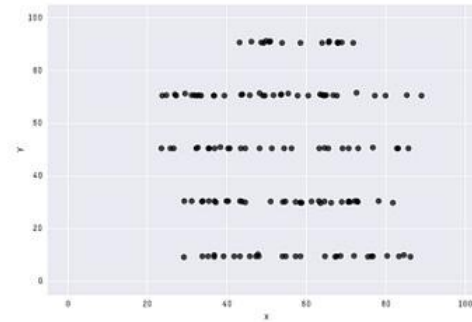
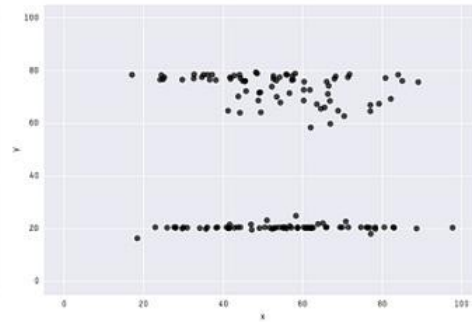
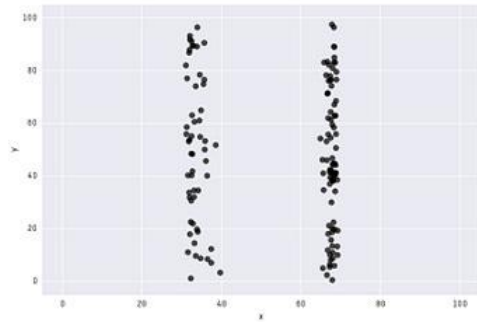
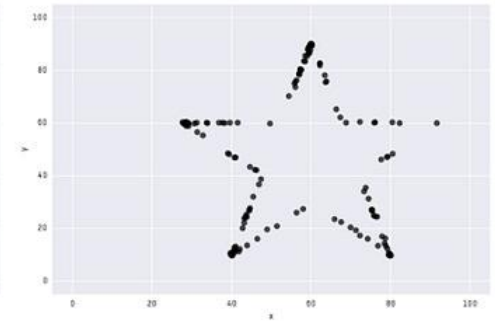
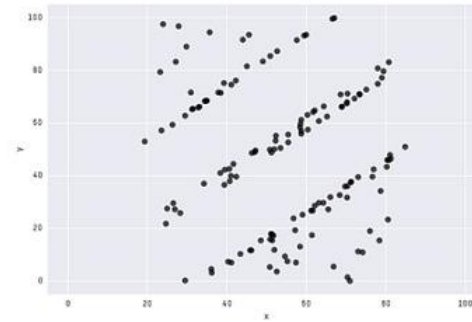
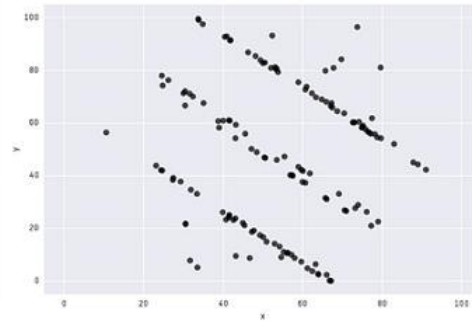
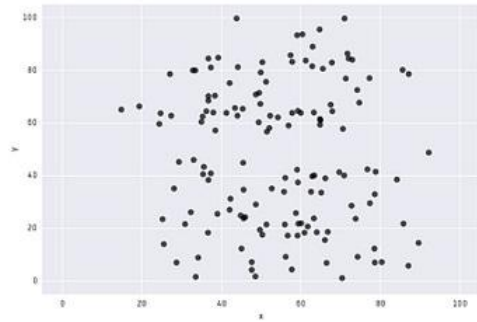
d	$P(d)$	Relative size of $P(d)$
1	30.1%	
2	17.6%	
3	12.5%	
4	9.7%	
5	7.9%	
6	6.7%	
7	5.8%	
8	5.1%	
9	4.6%	

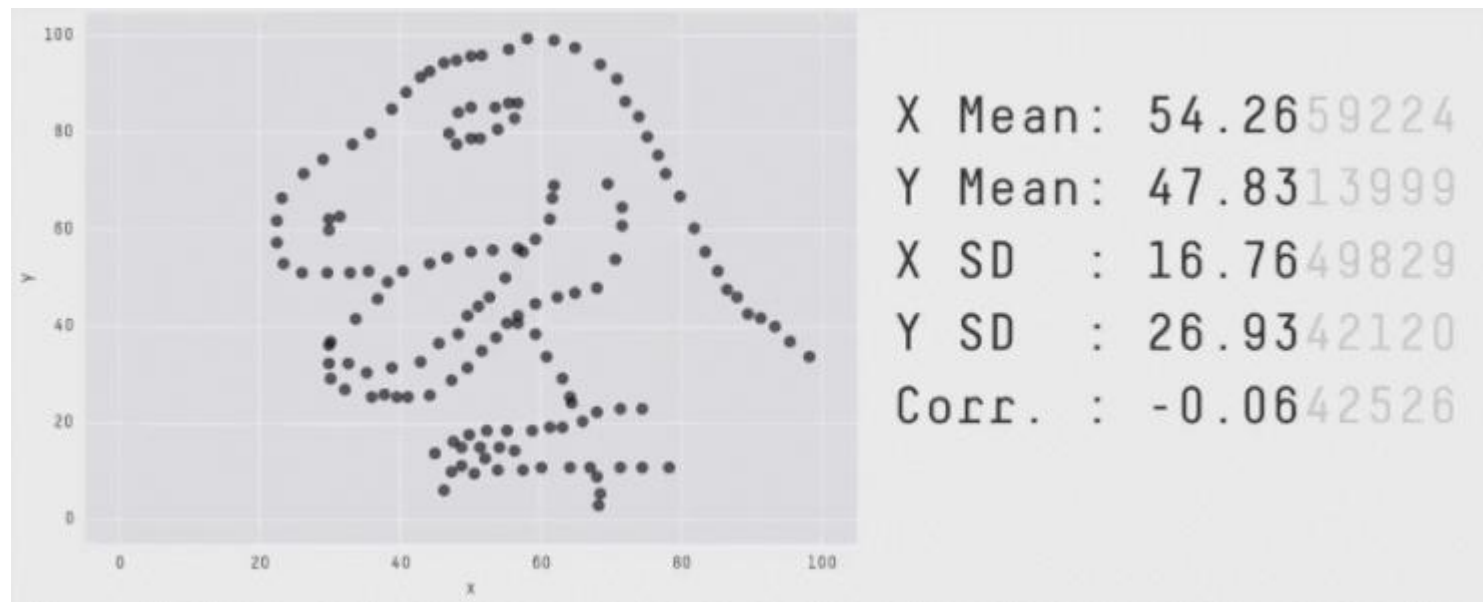




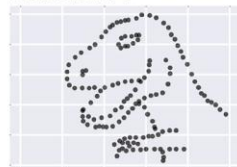


X Mean: 54.26
Y Mean: 47.83
X SD : 16.76
Y SD : 26.93
Corr. : -0.06

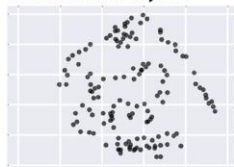




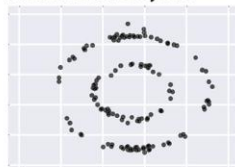
Iteration: 1



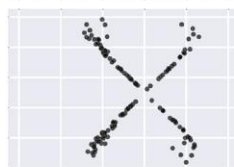
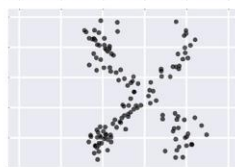
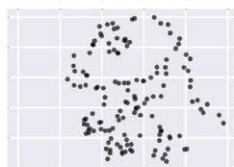
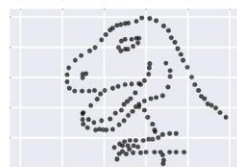
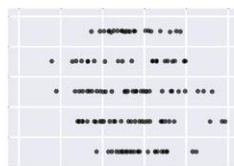
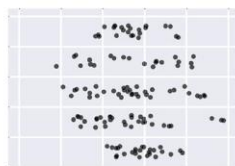
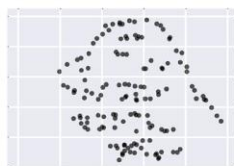
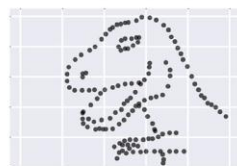
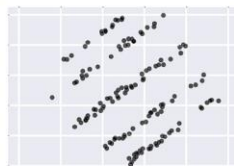
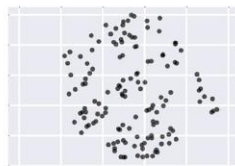
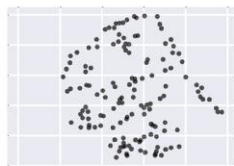
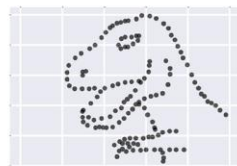
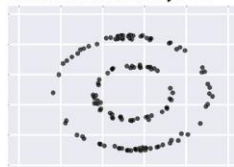
Iteration: 20,000

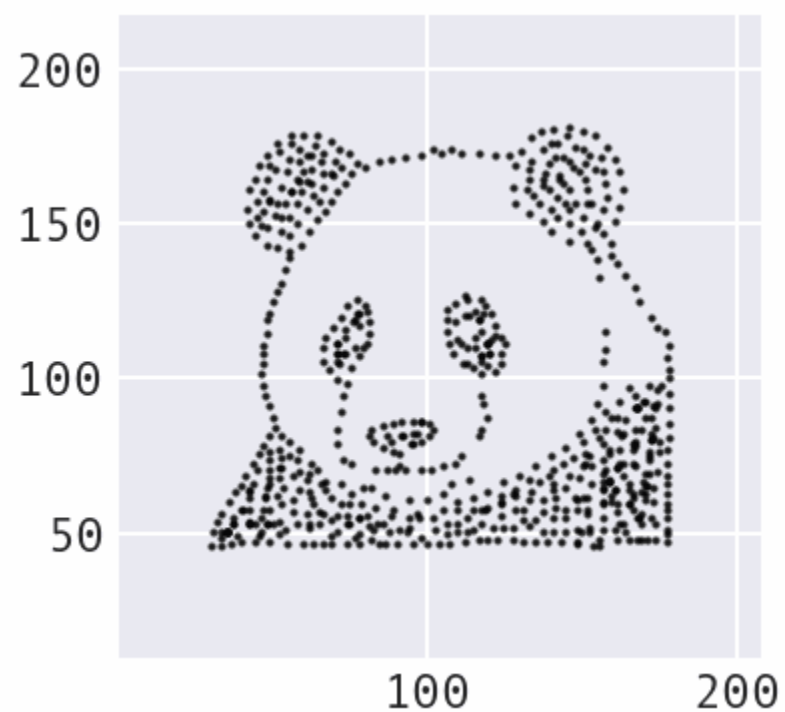


Iteration: 80,000



Iteration: 200,000





X Mean: 108.5222932

Y Mean: 95.6538752

X SD : 45.1183577

Y SD : 42.8928158

Corr. : -0.0522287